HATCHSIDE · PROBLEM STATEMENT

HatchSide problem statement

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DOCUMENT CONTROL

Document ID	HTS-PRB-001	Version	0.4
Status	Draft	Date	2026-11-05
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/hatchside		
Source	docs/01-problem.md @ 4004f47		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-11-05	Amish Chadha	Kit 1.9.1 upgrade (stage P3); users-and-needs and stakeholder tables added (feed HTS-URS-001)
0.3	2026-10-03	Amish Chadha	TRL 3; safety section added; first candidate partners recorded
0.2	2026-10-03	Amish Chadha	Populated to TRL 2; budget worded as a value-engineering target; open questions answered (HTS-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

HatchSide problem statement

Cleaning, sampling and retrieval jobs in sewers, tanks and digesters are still done by sending a person inside. The air in these spaces can kill in seconds, and it kills the rescuers too.

The problem

Deaths follow a pattern: one worker enters to clean or free something, collapses, and others follow. In a 1989 Michigan case, five family members died one after another in a manure pit (NIOSH FACE 89-46). In India, most sewer and septic tank victims audited had no safety equipment (The Wire, 2025), and in the US repeated tank cleaning deaths followed a failure to test the atmosphere before entry (OSHA, 2024).

Existing tools cover parts of the job. The Bandicoot robot cleans manholes with a pneumatic arm, but beta units cost about INR 10 to 12 lakh to make (Forbes India). Commercial confined-space tripods with winches cost about USD 5,850 to 6,120 and are built to lower and retrieve people (PK Safety). Sludge samplers exist but work alone (Hach). What is missing is a low-cost, open frame that carries many non-entry tools, so that the default for routine jobs becomes staying outside.

Users and context

Table 1. Users and needs (desk research, to confirm with co-design partners; the needs feed HTS-URS-001).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Sanitation and septic tank workers	Clear a blockage, take a sample or recover a dropped item without going down; kit they can carry on foot or by handcart	A crew of two or three at a street manhole or household septic tank, often informal work with little or no safety gear	No gas detection or rescue gear; cost of any kit; heavy loads on narrow lanes; tools that need training
Municipal and utility crews	Fast set-up on routine desilting and inspection; one kit for several jobs; kit that survives daily use	Sewer network, pump station and drain rounds from a truck, several sites a day	Traffic at the opening; short job windows; maintenance by their own workshop
Farm families and biogas technicians	Service and reseal a household fixed-dome digester from the opening, without entering a dome that holds methane and hydrogen sulfide	A technician with one helper at a 0.5 to 0.6 m dome opening, on foot or by motorbike	Small opening; digester up to about 2.5 m deep; no workshop; very low budget
Oil, gas and industrial tank contractors	Sample and loosen residues from the hatch before any entry decision	A crew at a storage tank roof hatch or frac tank, with permits and gas monitors already in place	Flammable atmospheres; hatch height above ground; rules on equipment used at the hatch
Safety and rescue teams (secondary)	A way to clip a line to a casualty from the rim without entering, with the haul made on certified rescue gear	A standby team at a permit-required space with its own rescue equipment	Time; kit that must not be mistaken for certified rescue equipment

Stakeholders

Table 2. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Sanitation workers and their organisations	Users and the people most at risk; they decide whether the kit is used	Co-design partner (none agreed; HTS-DDR-001, D11); walk through each head at a mock opening
Municipal sewer and sanitation departments	Buyers and operators of routine crews; responsible under the 2013 Indian act and the mechanised sanitation programme	Candidate partner (none agreed); confirm opening sizes, depths and routines
Household biogas programmes and technicians	Users of DomeReach; programme funders	Candidate partner (none agreed); confirm digester openings and sealant practice
Industrial tank operators and contractors	Users under permit-required confined space rules	Review of the retrieval head and equipment rules before any use claim
Regulators and standards bodies (labour, machinery, traffic)	Set the rules the kit must not be mistaken for breaking (lifting equipment, personnel rescue, confined space entry)	Read HTS-REG-001; ask a regulator or notified body before any sale or claim
Local fabricators and galvanizers	Make and mend the kit; they decide whether it can be built locally	Review the drawings and the build plan (BLD-001) for buildability
Traffic and public at the opening	Bystanders and passing vehicles at street manholes	Rim guard, cones and barrier chain; local work-zone rules to assess
Amish Chadha (Design Molecule)	Owner; decides scope, budget and release	Decisions listed in HTS-DEC-001 and docs/REVIEW.md
Funders and open hardware community	Fund trials; reuse and improve the design	Open licence (CERN-OHL-S-2.0); public repository after Amish's release decision

Operating environment

- Street manholes about 0.6 m (24 in) across and tank hatches on roofs or sides; depths to about 10 m (33 ft) assumed for a first version (estimate).
- Toxic, flammable or oxygen-deficient atmospheres inside the space, including hydrogen sulfide, methane and carbon monoxide.
- Traffic on streets, uneven ground, mud and sewage splash at the rim.
- Outdoor use in heat and monsoon rain; corrosive sewage and sludge on every wetted part.
- Crews of two to four people, with a handcart or small vehicle for transport.

Constraints

- Value-engineering target of USD 5,000 for the tripod, winch and the five tool heads including DomeReach (a hypothetical control target, not a spending limit).
- Hand winch only; no engine at the hatch.
- Kit carried in loads of 25 kg (55 lb) or less each (target).
- Every tool head worked from outside the space; no design step may require entry.
- Open design: hardware under CERN-OHL-S-2.0, any software under MIT.

Out of scope

- Planned human entry and the equipment that supports it (breathing apparatus, entry permits).
- Powered robots and articulated robotic arms.
- Gas detection electronics, which crews should buy as certified instruments.
- Sewer jetting and vacuum trucks.
- Building or repairing digester structures beyond sealing from the opening.

Prior work

Prior work	What it does	Gap for these users	Source
Genrobotics Bandicoot	Pneumatic, remote-controlled manhole cleaning robot with a sweeping arm	High cost and a powered robot; the preliminary screen notes three patents and ongoing litigation around its arm and gripper	link
3M DBI-SALA confined space tripod and winch	Aluminium tripod of 2.13 or 2.74 m with a retrieval winch for entry and rescue	Built for lowering people in; costs about USD 5,850 or more and carries no work tools	link
US5431248A confined space lowering and retrieving apparatus	Tripod or quadruped frame for lowering and retrieving personnel through manways	Expired patent for a personnel system; free prior art for a frame	link
Sludge Judge sampler	Core sampler for settled solids depth in tanks	Single task and hand-held; limited depth and no frame	link
Biogas plant maintenance guidance	Guidance that digester maintenance should be done from above ground where possible	Guidance only; no tools for small household digesters	link

Co-design

A sanitation workers' organisation or municipal sanitation department that already runs mechanised cleaning programmes, plus a biogas programme with field technicians for DomeReach, so heads are shaped by the real jobs and real openings.

First candidates to approach (none agreed): a municipal sewer department in India running mechanised cleaning under the national mechanised sanitation programme, and a household biogas programme with field technicians in India or Nepal (HTS-DDR-001, D11).

Co-design checklist:

- [ ] Confirm the five tasks with crews who do them today.
- [ ] Measure common manhole, tank hatch and digester openings in the first region.
- [ ] Walk through each head at a mock opening with the crew.
- [ ] Confirm that crews will carry a certified gas detector.

Safety

> **Safety:** HatchSide lifts loads over an open confined space whose air can kill in seconds. It is an open engineering reference for a material-handling frame, never certified rescue, lifting or fall-protection equipment. Nobody enters the space, and the winch never lifts or lowers a person. Test the air at the opening with a certified gas detector before and during work, keep sparks and flames away from sewers and digesters, guard the opening from traffic and falls, and never stand under a raised load.

Answered questions

The scaffold's open questions were answered at the TRL 2 review (HTS-DDR-001) and are tracked in the design decisions register (HTS-DEC-001):

- Tasks that still force entry: none of the five routine tasks needs entry with the five heads; any task found in co-design that does gets a new head, not an entry.
- The retrieval head stays, as a clip-on guide only; the haul is made with certified rescue equipment.
- The frame carries 150 kg; WellSling's 400 to 600 kg loads need their own frame.
- DomeReach uses a generic roller-applied digester sealant, named by type.
- First trial hosts are recorded as candidates to approach, above.